

Contextual Refinement of Translations: Large Language Models for Sentence and Document-Level Post-Editing

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Abstract

Large language models (LLMs) have demonstrated considerable success in various natural language processing tasks, but open-source LLMs have yet to attain state-of-the-art performance in Neural Machine Translation (NMT). Nevertheless, their significant performance in tasks demanding a broad understanding and contextual processing shows their potential for translation. To exploit these abilities, we investigate using LLMs for MT and explore recent parameter-efficient fine-tuning techniques. Surprisingly, our initial experiments found that fine-tuning with Q-LoRA for translation purposes led to performance improvements in terms of BLEU but degradation in COMET compared to in-context learning. To overcome this, we propose an alternative approach: adapting LLMs as Automatic Post-Editors (APE) rather than direct translators. Building on the ability of the LLM to handle long sequences, we also propose extending our approach to document-level translation. We show that leveraging Low-Rank-Adapter fine-tuning for APE can yield significant improvements across both sentence and document-level metrics while generalizing to out-of-domain data. Most notably, we achieve a state-of-the-art accuracy rate of 88.7% on the ContraPro test set, which assesses the model’s ability to resolve pronoun ambiguities when translating from English to German. Lastly, during manual post-editing for document-level translation, the source sentences are iteratively annotated, which can be used to refine further translations in the document. Here, we demonstrate that leveraging human corrections can significantly reduce the number of edits required for subsequent translations.

1 Introduction

Large Language Models (LLMs) are currently being explored for many Natural Language Processing tasks such as Question Answering and Dialogue Applications (Touvron et al., 2023; Anil et al.,

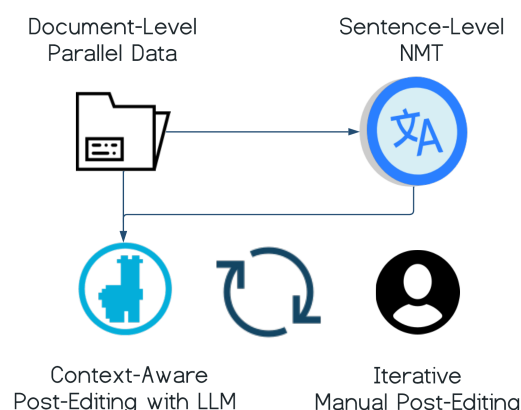


Figure 1: **Iterative Manual Post-Editing:** For manual PE, the annotator supplies the gold target context iteratively. The LLM then utilizes this gold target context for generating translations at document-level.

2023; Thoppilan et al., 2022; Tan et al., 2023). Moreover, they are even shown to achieve or surpass state-of-the-art performance based on traditional methods. This achievement demonstrates their ability to possess general understanding and process long inputs. Given these strengths, LLMs might also be suitable for Machine Translation (MT) as many of them are also inherently multilingual from being trained on the web.

However, LLMs for MT still remain an under-explored area of research. While there are initial works on using LLMs for MT at both sentence and document-level (Vilar et al., 2022; Hendy et al., 2023; Wang et al., 2023; Zhang et al., 2023), the performance of open-source models in the range of 13B parameters still lags behind the current state-of-the-art Neural MT (NMT) methods (Xu et al., 2023). Nonetheless, closed models such as GPT-4 are on par or better depending on the language pair (Kocmi et al., 2023). Hence, it is necessary to investigate adapting small-scale open-source models for translation.

Notably, these methods mainly employ In-Context-Learning (ICL) (Brown et al., 2020) as fine-tuning these models often requires a significant amount of computational resources. Hence, there might be a barrier to optimally adapting the LLMs for MT.

Parameter-efficient techniques for fine-tuning such as Low-Rank Adapters (LoRA) (Hu et al., 2021; Dettmers et al., 2023) were recently proposed to overcome large computational requirements. This enables a new adaptation process for LLMs. However, whether these techniques are sufficient for successful adaptation and better generalization is still unclear.

This work investigates exploiting LLMs for MT at both sentence and document levels. We initially experiment using ICL and parameter-efficient fine-tuning techniques such as Q-LoRA to use LLM for MT and find that adding adapters alone is insufficient and may even lead to degradation in COMET (Section 4). While full fine-tuning may resolve such discrepancies, it is important to efficiently adapt open-source LLMs with minimal computational resources. To mitigate this and still exploit the strengths of LLMs, we propose to adapt them as Automatic Post-Editors (APE) correcting NMT systems hypothesis rather than direct translators.

This approach offers several advantages. It introduces modularity, allowing state-of-the-art or customized NMT techniques to be applied independently, followed by LLM improvements. Additionally, LLMs can refine the sentence-level NMT systems output and generate consistent and coherent text using their ability to generate fluent and long documents.

The cascaded system of NMT and LLM offers modularity and can also enable the integration of human feedback. By feeding the LLM human-corrected translations from previous sentences in the document, we show that it can leverage this feedback to improve the current sentence’s translation. This process can be applied iteratively in practice, sentence by sentence, as the annotator progressively corrects the translated document (see Figure 1 for Iterative Manual Post-Editing).

We summarize our main findings and contributions below:

- **Effective Combination of NMT and LLM:** Our sentence-level LLM APE demonstrates a successful fusion of knowledge from NMT systems and LLMs, leading to substantial en-

hancements in translation quality. Importantly, we observe that the LLM APE exhibits robustness and can adeptly correct NMT systems, even for test sets from different domains that it was not explicitly trained on.

- **Extension to Document-level Post-Editing:** We extend our approach to document-level APE and observe significant improvements in both sentence and document-level translation metrics. Notably, we achieve a state-of-the-art accuracy of 88.7% on the ContraPro English to German test set, underscoring the effectiveness of APE.
- **Iterative Manual Post-Editing:** We introduce a promising use-case scenario for iterative post-editing (as depicted in Figure 1). We show that providing gold target context significantly enhances the remaining translation quality, both at the sentence and document levels, as indicated by various metrics.

2 Approach: Adapting LLM for APE

Open-source LLMs may not be as proficient translators as state-of-the-art NMT systems due to no explicit training with large amounts of parallel data. However, LLMs being trained on the web containing data from several domains possess general knowledge that is lacking in an NMT model. Moreover, they are capable of processing significantly longer inputs compared to a standard sentence-level NMT. Given their strengths of knowledge and ability to process long sequences, we propose to use them for APE at both sentence and document levels. Hence, we first generate translations with NMT and then perform APE with LLMs. Our approach combines the translation capacity of NMT with fluency and understanding of LLMs.

We use a technique similar to Niehues et al. (2016), which combines Phrase-Based and NMT models. We extend this approach to incorporate LLMs for APE at both sentence and document levels. We first explain our system’s complete pipeline at both text representation levels of sentence and document. Then, we describe how we train and create the data for each step in our setup. Finally, we explain how human feedback can be easily integrated into our cascaded approach during manual PE.

2.1 Pipeline

Given a source sentence s , we use an NMT model to generate an initial translation h_{NMT} . Then for our APE model, we do not provide the h_{NMT} alone as it cannot distinguish when the hypothesis from NMT is severely mistranslated but still fluent. Hence, we feed the source sentence and the initial translation to LLM and generate a refined hypothesis h_{LLM} .

$$h_{NMT} = \mathcal{G}(\theta_{NMT}, s) \quad (1)$$

$$h_{LLM} = \mathcal{G}(\theta_{LLM}, s, h_{NMT}) \quad (2)$$

where θ_{NMT} and θ_{LLM} are models trained for translation and APE. $\mathcal{G}(\theta, s, h)$ indicates generating a hypothesis using the model θ given s and h using beam search.

For APE at the document level, we extend the above formulation to process a sequence of sentences. Consider a document $\mathcal{D}$ with n source sentences s^i where i ranges from 1 to n . We first use the NMT model to generate each translation in isolation at the sentence level. Let them be denoted as h_{NMT}^i . Then, we perform APE using the sequence of source and hypothesis sentences, exploiting the LLM’s ability to process and use contexts. We denote the generated document translation as h_{LLM}^D :

$$h_{NMT}^i = \mathcal{G}(\theta_{NMT}, s^i) \forall i \in 1..n \quad (3)$$

$$h_{LLM}^D = \mathcal{G}(\theta_{LLM}, s^D, h_{NMT}^D) \quad (4)$$

where s^D and h_{NMT}^D are the source and sentence-level hypothesis sentences joined by a separator token to form a document.

2.2 LLM Fine-tuning for APE

In our cascaded approach, we have an NMT and an LLM. For training our NMT model, we train it on available parallel data in a conventional fashion. We do not take any additional steps as our main motivation was to exploit LLMs for further enhancements. In the case of the LLM, we propose to go beyond ICL approaches and fine-tune them for maximum utility, as described in the following.

2.2.1 Training on MT Errors

To further optimize the LLM for the task of APE, we propose to fine-tune them using Q-LoRA (Hu et al., 2021; Dettmers et al., 2023). It is ideal to fine-tune the LLM by providing the source and hypothesis as input and predicting the corresponding

post-edited reference. This needs data in the form of triples comprising the source, initial hypothesis, and reference. To simulate real test conditions, we need the initial hypothesis to consist of the errors generated from the NMT model we plan to use.

To achieve this, we follow these steps:

1. We partition the training data into two halves.
2. We train two separate NMT models, one for each half of the data.
3. We utilize the model trained on the first half to perform inference on the second half, and vice versa.

This process yields the same quantity of instances as the original training set, with initial hypotheses that exhibit errors typical of the NMT system we plan to improve on. Subsequently, we format this data into a prompt template, as described in Appendix A.4 or A.5, depending on the level of representation. Then, we employ Q-LoRA for fine-tuning¹ our LLM. For document-level APE (Doc APE), we simply split into non-overlapping chunks according to a chunk limit of source tokens and create our training data.

2.2.2 Inference

In our setup, the level of granularity can be a sentence or a document. The decoding process is straightforward for sentence-level APE (Sent APE). We generate an initial translation and feed it to the adapted LLM for our final refined hypothesis.

In the context of document translation, decoding poses a more intricate challenge compared to sentence-level. Decisions must be made regarding the direction of context for each source sentence, whether it should be drawn from the left, right, or both sides. In our work, we explore the following strategies:

Chunk-Based: This is a straightforward approach where we employ the same method used to create our training data. We create non-overlapping chunks and translate them individually. In this setup, it’s possible that some sentences may lack left or right context. Note that if the number of sentences in the hypothesis doesn’t match the source, we replace them with the sentence-level Δ LM outputs exploiting the modularity of the cascaded approach. We’ve observed that this situation occurs

¹Training details can be found in Appendix C.1

infrequently, with at most 30 sentences, and thus for rare instances.

Batched Sliding Window: We translate the document using a sliding window approach with a payload, as described in [Post and Junczys-Dowmunt \(2023\)](#). Following our chunk limit, we append the sentence we intend to translate with as much preceding source context as possible. Then, we translate the entire chunk, including the context (*Payload*), and extract the last sentence using the separator token.

Continuous Sliding Window: Similar to the previous approach above, we append the left source context according to the chunk limit for translation. However, the key distinction here lies in not regenerating the target context at every step. When translating a sentence, we force-decode the previous sentence translation that is the target context in the next step. Hence, at each step, only one sentence is translated into the target language, which is then used for forced decoding in the subsequent step to provide the target context (Referred to as *Sequential Decoding* in [Herold and Ney \(2023\)](#)).

2.3 Integrating Manual Feedback

Consider the case of manual PE, where the annotator edits each sentence in the document. Here, we have access to the human-corrected target context, which can be used to refine future translations.

We propose to integrate this information into our APE system. We condition its subsequent translations on this expert knowledge by iteratively feeding the model human-corrected contextual information from preceding sentences and appending it to the prompt. This modular approach enables straightforward integration of human input without requiring additional training.

3 Experimental Setup

Models: In our proposed approach, we have a sentence-level NMT system that generates an initial hypothesis and an LLM which then improves it. Nonetheless, we want a strong NMT model to assess the benefits of using LLMs. Furthermore, the model should be efficient to add more latency to the two step approach. Therefore, we fine-tune the pre-trained DeltaLM² ([Ma et al., 2021](#)) (Δ LM) for initializing our NMT sentence-level model (Refer to Appendix C.2 for more details). We chose this

model given its size and the performance in the constrained setup in the IWSLT evaluation for multilingual track ([Agarwal et al., 2023](#)). We also ablate with NLLB ([Costa-jussà et al., 2022](#)) to compare with state-of-the-art NMT models (Section 5.3). While For LLM, we use the recently open-sourced Llama-2-13b-chat-hf ([Touvron et al., 2023](#)) as it is instruction-finetuned and has reasonable compute and memory requirements when adapting with 4bit Q-LoRA ([Dettmers et al., 2023](#)).

During training (Refer to Appendix C.1 for more details), we mask the loss on the prompt, which means that the LLM is exclusively trained to predict the reference given the source and hypothesis.

Datasets & Metrics: We primarily focus on translating talks from **English to German** at a document-level. This choice is based on the large availability of document-level parallel data, the current state of sentence-level NMT quality, and the necessity for contextual information in this translation direction.

For training our sentence NMT and post-editor LLMs, we utilize the MuST-C V3 Corpus ([Di Gangi et al., 2019](#)). This corpus aligns well with our objectives, containing parallel data annotated with talk IDs for document-level translation.

For testing, we report results on three test sets. First, we select a subset of the training corpus with the most contextual phenomena in the speeches. This choice stems from the need for context not always prevalent, and hence, standard tests may not suffice for document-level evaluation. To identify such contextual phenomena, we employ the MuDA tagger ([Fernandes et al., 2023](#)), which automatically tags words requiring contextual information in the training corpus. We select talks with the highest number of tags for each phenomenon, resulting in 14 talks in our test sets, addressing contextual occurrences related to pronouns, formality, and lexical cohesion. Then, we remove the selected talks from our training data and use them for testing.

To evaluate the robustness of our approach with out-of-domain data, we use the WMT21 News test set ([Akhbardeh et al., 2021](#)) and the ACL dev set from IWSLT23 ([Salesky et al., 2023](#)). Although the ACL data consists of talks, its content contains terminology and domains that are unlikely to be found in the training data. Furthermore, both test sets are annotated at the document level, aligning with our experimental setup.

Table 1 presents an overview of the datasets. Notably, the number of detected tags is relatively small

²We use the Δ LM base model with 360M parameters

Dataset	Sentences	Documents	MuDA Tags		
			Pronouns	Formality	Lexical Cohesion
MuST-C V3 Train	261.4K	2.5K	28K	82K	86K
MuST-C V3 Test	3637	14	332	1127	1268
WMT 21 News	1002	68	42	145	381
ACL Dev	468	5	12	38	478

Table 1: Statistics of our training and test data sets. We report the number of sentences and documents along with the total tag occurrences annotated by the MuDA tagger.

compared to the total sentences, even when accounting for false positives in WMT and ACL test sets. Therefore, creating a custom test set was essential to reliably evaluate context usage and sentence-level translation quality.

We also report scores on the ContraPro (EN $\rightarrow$ DE) (Müller et al., 2018) for a targeted evaluation of context usage in resolving pronoun ambiguities.

Regarding metrics, we employ various methods to assess the quality at both the sentence and document levels. Our report includes BLEU (Papineni et al., 2002), ChrF2 (Popović, 2016) using SacreBLEU (Post, 2018), and COMET³ (Rei et al., 2022) scores for sentence-level evaluation. To gauge the quality of word prediction using contextual information at the document level, we report Precision, Recall, and F1 scores for words detected by the MuDA tagger.

4 Automatic Post-Editing is Necessary

Shot Size	BLEU	ChrF2	COMET
Sent-level Δ LM	30.45	57.0	0.8179
Llama2 13B ICL (Random 2)	21.53	50.0	0.7795
Llama2 13B MT	28.92	55.9	0.7664
Llama2 13B 0-Shot APE	27.26	53.9	0.8008

Table 2: ICL, LoRA fine-tuning for MT and 0-Shot APE performance of the Llama2 13B on Must-C test set. We report BLEU, ChrF2 and COMET scores for each configuration (EN $\rightarrow$ DE) and highlight the best scores in **bold** for each metric. We report only best performing ICL configuration but provide results of 1-5 shots in the Appendix B.

While APE with LLMs seems intuitively advantageous, our first step is to empirically evaluate it against several baselines, including alternatives like In-Context-Learning (ICL) (Brown et al., 2020)

and fine-tuning with LoRA. To justify the development of a cascaded system with added computational complexity, we assess the following configurations⁴:

Sentence-Level NMT: We fine-tune Δ LM on the training data at sentence-level in conventional fashion.

In-Context-Learning with Llama2 13B: We prompt the model according to Vilar et al. (2023), selecting examples randomly or similar to the current prompt. To find sentences more closely related to our source, we extract sentence embeddings from our training data using Sent-BERT (Reimers and Gurevych, 2019) and retrieve the nearest neighbors for efficiency using FAISS (Johnson et al., 2019) (Llama2 13B ICL).

Llama2 13B + Adapters: Leveraging the recent advancements in efficient fine-tuning of LLMs, we fine-tune with adapters using LoRA (Hu et al., 2021) (Training and Hyper-Parameter Details can be found in Appendix C.1). Like the sentence-level NMT, we fine-tune it on all of our training data at the sentence level (Llama2 13B MT).

Zero-Shot Post-Editing with Llama2 13B: Finally, we consider the case of simple zero-shot PE to evaluate the in-built ability of the model to use the knowledge from another system and compare it to ICL where it acts as a direct translator.

Results for the above setups are reported in Table 2. First and foremost, we observe that the sentence-level Δ LM achieves the highest scores across all metrics. This underscores the highly competitive performance of a dedicated NMT model with 360M parameters compared to a 13B LLM.

Furthermore, we find that in the case of ICL, the selection strategy is relatively unimportant, with both random and FAISS performing similarly as indicated by the scores in Appendix B. Addition-

³Unbabel/wmt22-comet-da

⁴Prompt templates for all the configurations described in Appendix A

ally, increasing the number of exemplars in the prompt had a detrimental effect on our setup. Moreover, adapting with LoRA yields the highest BLEU and ChrF2 scores of 28.92 and 55.9 when compared to setups that rely solely on the LLM. However, it’s worth noting that COMET scores decrease compared to ICL. These findings suggest that fine-tuning with Q-LoRA on extensive parallel data may lead to higher scores in lexical metrics but degradation in COMET. Whether such a trend is also observed with full fine-tuning demands further exploration, and we do not perform it due to a lack of computational resources.

Zero-Shot APE beats ICL across metrics (COMET included), unlike LoRA, showing LLMs’ innate post-editing ability. However, it falls short of sentence-level Δ LM. Therefore, we propose to train the adapter for APE rather than direct translators to exploit LLMs.

5 Llama2 13B as Sentence-Level Post Editors

Model	BLEU	ChrF2	COMET
<i>MuST-C V3</i>			
Δ LM	30.45	57	0.8179
Llama2 13B MT	28.92	55.9	0.7663
Llama2 13B 0-Shot APE	27.26	53.9	0.8009
Δ LM + Llama2 13B Sent APE	31.71	58.3	0.8330
<i>WMT 21 News</i>			
Δ LM	21.53	52.6	0.7911
Llama2 13B MT	23.61	54.3	0.7931
Llama2 13B 0-Shot APE	21.44	52.0	0.7982
Δ LM + Llama2 13B Sent APE	25.16	56	0.8411
<i>ACL Dev</i>			
Δ LM	31.36	60.5	0.7945
Llama2 13B MT	31.47	60.5	0.772
Llama2 13B 0-Shot APE	30.83	60.3	0.8028
Δ LM + Llama2 13B Sent APE	36	63.9	0.8321

Table 3: Performance of Sent-Level Llama2 13B APE on test sets in and out of the domain. Δ LM + Llama2 13B Sent APE denotes using the hypothesis of Δ LM as input for our adapted Sentence-Level Llama2 13B post editor. We report BLEU, ChrF2, and COMET scores for each approach (EN $\rightarrow$ DE) and highlight the best scores in **bold** for each metric.

In this section, we evaluate the performance of improving sentence translations with APE. First, we discuss the results of improving translations generated only by Δ LM on in-domain test data. Then, we analyze the influence of moving away from our training conditions to assess the robustness of the

model. We achieve this by first evaluating its performance on out-of-domain test sets and combining it with hypotheses generated by models other than Δ LM.

5.1 Improved Translation Quality with Sentence-Level APE

We evaluate our Sentence-Level Llama2 APE and present the results in Table 3. To assess the utility of APE, we also report scores for the individual models, namely, the sentence-level Δ LM and the Llama2 fine-tuned with LoRA on the parallel data.

We see that post-editing the output of Δ LM with Llama2 outperforms other models across all metrics while fine-tuning for MT alone shows degradation. We hypothesize that this is primarily due to LLMs’ internal knowledge and intrinsic ability to generate fluent sentences while lacking in translation capability. However, by providing initial translations to make the task easier, LLM improves the quality by a high margin.

5.2 Generalizability to Out-Of-Domain Data

From Table 3, we observe that the performance gains are more pronounced on the WMT21 News and ACL test sets compared to our MuST-C test set. The primary difference is that WMT and ACL fall outside the domain of the training data. Hence, we observe more significant improvements compared to our MuST-C test set. We gain by 1.26 BLEU on MuST-C but up to 5.64 and 3.63 BLEU on the out-of-domain ACL and WMT test sets respectively.

This scenario mirrors practical situations where a system encounters out-of-domain sentences and performs sub-optimally. By utilizing Llama2, containing a broader spectrum of "knowledge" (Illustrated in Table 10), we demonstrate that it can significantly enhance translation quality.

5.3 Generalizability to NMT Models

Apart from improving Δ LM hypothesis, it is ideal if the APE with Llama2 can enhance translations of various NMT models. Therefore, to critically assess the generalization ability of the Sentence-Level Llama2 APE, we evaluate it on correcting hypotheses that were not generated from Δ LM on MustC and out-of-domain ACL dev set. For this purpose, we utilize the NLLB models⁵ (Costa-jussà et al., 2022) and present the results in Table 4.

⁵We perform inference with 8-bit quantization and achieve slightly lower scores than reported in the literature

Model	BLEU	ChrF2	COMET
<i>MuST-C V3</i>			
Llama2 13B MT	28.92	55.9	0.7663
NLLB 3.3B	31.6	58.8	0.8265
NLLB 3.3B + Llama2 13B Sent APE	32.44*	58.9*	0.8320*
<i>ACL Dev</i>			
Llama2 13B MT	31.47	60.5	0.772
NLLB 3.3B	43.01	69.7	0.8321
NLLB 3.3B + Llama2 13B Sent APE	40.09	67.2	0.8372*
NLLB 54B	45.82	71.56	0.844
NLLB 54B + Llama2 13B Sent APE	40.91	67.8	0.8407

Table 4: Analyzing the robustness of the Llama2 13B Sentence-Level APE. *NLLB X LLM + LLM Sent APE* denotes using the hypothesis of NLLB X as input for our adapted Sentence-Level LLM APE. Best scores for a test set are in **bold** for each approach. If the post-editor improves the hypothesis according to a metric, we denote it with *

We show that for the in-domain test set MuST-C, the APE successfully improves the scores in all metrics with notably an increase in COMET score of 0.65. In the case of out-of-domain ACL, APE improves the COMET score for the 3.3B NLLB model (0.5 gain) but hurts lexical metrics, suggesting it rephrases outputs while maintaining quality. However, APE harms 54B NLLB translations, likely due to difficulty finding errors in such strong models and adapter training focused on lower-quality hypotheses.

6 Llama2 as Document-Level Post Editors

Another motivation for our approach is to exploit LLMs ability to process long sequences for Doc APE. In this section, we evaluate and analyze the performance of our Doc APE model in detail.

To gain insights on whether the Doc APE with Llama2 is beneficial, we compare it against several models. Apart from the previously mentioned sentence-level models such as $\triangle$ LM and Llama2 + LoRA, we also extend them to the document-level by concatenating sentences (Tiedemann and Scherrer, 2017) (*Doc2Doc*). Furthermore, we evaluate different decoding strategies and report both sentence and document-level metrics in Table 5.

After tuning on the dev data, we set the Llama2 maximum chunk token sizes as 1024 for training and 256 for inference (See Figure 2 for more information). This ensured at least 5 preceding sentences for most data, which we found to be reasonable given the computation requirements with

large inputs. For $\triangle$ LM Doc2Doc, we use a smaller chunk size (128 tokens) due to its limited capacity.

Concatenating Sentences for Doc2Doc Proves Insufficient: Our analysis reveals that models fine-tuned with $\triangle$ LM and Llama2 separately at the document level exhibit subpar performance when compared to the sentence-level $\triangle$ LM across all considered metrics. This limitation likely stems from the scarcity of document-level parallel data, a common occurrence, particularly in the context of low-resource languages. This highlights the inadequacy of concatenation as a standalone approach in practical use cases.

Navigating the Trade-off between Sentence and Document APE: Doc APE models outperform sentence-level on BLEU/ChrF2 (despite slight COMET dip of 0.003 between Sent and Doc APE), showing promise for document translation. The Doc APE models leverage context effectively for pronouns and formality by achieving the best F1 scores of 0.75 and 0.74. However, it is still unclear why the COMET score of Doc APE model is worse while we observe improvements in all other metrics.

Impact of Decoding Strategy: Doc APE’s different decoding strategies (chunking, windowing) show no clear winner in the sentence or document-level metrics. Batched sliding window, though computationally expensive, offer no significant advantage. Thus, a continuous sliding window or chunking may be preferred for efficiency. However, further research across domains and languages is crucial to comprehensively understand Doc2Doc decoding strategies.

6.1 Incorporating Target Context during Manual Post Editing

Until now, we have focused on APE and assumed there is no human feedback. However, in the case of manual PE, we force decode the previous target sentences as the manually corrected target context and condition the model to generate the translation of the current source sentence. We denote this as $\triangle$ LM + Llama2 Doc APE Gold Target Context in Table 5.

By feeding gold target sentences as context to Doc APE, we achieve substantial gains across metrics: +4.14 BLEU, +2.6 ChrF2, +0.268 COMET, compared to sentence-level $\triangle$ LM. This validates Doc APE’s ability to leverage context and suggests the potential for reducing manual edits in PE, leading to cost savings.

Approach	BLEU	ChrF2	COMET	Pronouns			Formality			Lexical Cohesion		
				Precision	Recall	F1	Precision	Recall	F1	Precision	Recall	F1
Δ LM	30.45	57	0.8179	0.65	0.76	0.70	0.68	0.70	0.69	0.60	0.74	0.67
Δ LM Doc2Doc	30.66	57.7	0.7481	0.66	0.78	0.71	0.68	0.72	0.69	0.6	0.74	0.66
Llama2 13B MT	28.92	55.9	0.7663	0.66	0.77	0.71	0.67	0.71	0.69	0.61	0.76	0.68
Llama2 13B MT Doc2Doc	28.98	56.1	0.8221	0.67	0.75	0.71	0.68	0.74	0.71	0.61	0.70	0.65
Δ LM + Llama2 13B SENT APE	31.71	58.3	0.8330*	0.66	0.77	0.71	0.67	0.71	0.69	0.61	0.76	0.68*
Δ LM + Llama2 13B Doc APE Chunk	31.47	58.4	0.8306	0.68	0.82	0.74	0.66	0.76	0.71	0.60	0.76	0.67
Δ LM + Llama2 13B Doc APE Batched SW	31.77	58.9*	0.8300	0.68	0.83*	0.75*	0.67	0.77	0.72	0.61	0.77*	0.68*
Δ LM + Llama2 13B Doc APE Continuous SW	31.85*	58.9*	0.8298	0.69*	0.72	0.71	0.68*	0.81*	0.74*	0.62*	0.64	0.63
Δ LM + Llama2 13B Doc APE Gold Target Context	34.59	59.6	0.8347	0.73	0.8	0.76	0.77	0.78	0.77	0.69	0.77	0.73

Table 5: Comparing our Document Level APE with Llama2 13B with sentence level APE and conventional approaches. We use chunk-based decoding unless it is explicitly mentioned for Doc2Doc models. We report BLEU, ChrF2 and COMET scores for sentence level evaluation and MuDA tagger scores for document level. The best score in each metric is highlighted in **bold**. We also compare APE models without gold target context in isolation and append * for the best score in each metric.

6.2 Disambiguating Pronouns with Doc APE

We also report scores on the ContraPro test set (Müller et al., 2018). This is a benchmark designed to assess the disambiguation of pronouns, specifically "Er" (masculine), "Sie," (feminine) and "Es" (neutral) when translating "It" from English to German. We evaluate on two setups following Post and Junczys-Dowmunt (2023). For contrastive, we force-decode the target context and determine which pronoun is most likely based on the log-likelihood. In the case of generative, we directly translate the full source paragraph and extract the last sentence to check if it contains the correct pronoun.

	Cxt Size	Contra/Gen (%)
Post and Junczys-Dowmunt (2023)	10	77.9/ 70.5
Lupo et al. (2023)	4	82.54/_
Δ LM + Llama2 Doc APE	2	87.7/68.0
Δ LM + Llama2 Doc APE	4	88.7 /69.7

Table 6: Contrastive and Generative accuracy on the ContraPro English $\rightarrow$ German Test Set. Results for Sent APE and additional configurations in Table 9.

We find that our document-level APE model achieves state-of-the-art accuracy 88.7% in choosing the right pronoun. This can be attributed to LLMs pre-training in long texts. We are very comparable to Post and Junczys-Dowmunt (2023) for generative accuracy with fine-tuning only on TED talks. Moreover, this shows that when target context is made available, LLMs seem to better exploit them and are ideally suited for document-level.

7 Related Work

Document NMT: Conventional approaches in Doc-NMT rely on a straightforward concatenation technique (Tiedemann and Scherrer, 2017; Agrawal et al., 2018; Post and Junczys-Dowmunt, 2023). Several works also explored complicated adaptations to transformer architectures, such as the inclusion of additional context encoders (Jean et al., 2017; Voita et al., 2018), adjustments to positional information (Bao et al., 2021; Li et al., 2023), and the application of data augmentation strategies (Sun et al., 2022), among others. Similar to our work is Voita et al. (2019), where sentence-level translations are refined to create a coherent document but without considering the source context.

LLM for MT: LLMs are currently being explored for MT given their success in several tasks. These techniques were mainly facilitated by ICL (Brown et al., 2020) at sentence-level (Zhang et al., 2023; Vilar et al., 2022) or document-level (Hendy et al., 2023; Wang et al., 2023). Similar to our work, the other line of direction is integrating translation memories (Mu et al., 2023; Moslem et al., 2023) or correcting NMT system outputs in the prompt (Raunak et al., 2023; Chen et al., 2023). It is worth noting that our work sets itself apart from these approaches by leveraging efficient LoRA and enabling the effective fusion of NMT with LLMs at both the sentence and document-levels.

Online Learning for NMT: Integrating human feedback for MT was explored in both statistical and neural MT (Formiga et al., 2015; Logacheva,

2017). Many methods perform additional training steps using the feedback and alter the MT model at run-time (Turchi et al., 2017; Kothur et al., 2018). Few works explored integrating retrieval and cache mechanisms to avoid further fine-tuning (Gu et al., 2018; Shang et al., 2021; Wang et al., 2022). Our approach incorporates human feedback as context and does not need any changes.

8 Conclusion

Our work highlights LLMs’ potential for APE, significantly boosting NMT at both sentence and document levels. We showed that it enables modularity, deeper text understanding, and document-level quality boosted by LLMs’ massive pretraining.

For future work, we consider several research avenues. These include training the adapters on substantially larger volumes of document-level parallel data, assessing various open-source LLMs, and conducting similar experiments with low-resource languages and domain.

9 Limitations

The main disadvantage of the proposed cascaded system is the latency of generating a translation. From Table 5, we find the Δ LM performance is worse but comparable to the APE approaches with LLM. However, Δ LM can produce translations with significantly shorter latency than LLMs. Therefore, integrating techniques from quality estimation to decide when to perform APE may be helpful to overcome this limitation.

The other drawback of the cascaded approach is that it does not simulate a deep fusion. The LLM can make mistakes even when the NMT is highly confident and correct for a given translation. However, fusing them is not trivial due to the models having different vocabularies.

Finally, we would also like to mention that we performed experiments for only English to German direction, which was highly present during LLMs pretraining. The benefits of APE should also be validated for low-resource languages for generalizability, where the monolingual data of such languages may be significantly less in the LLM pretraining.

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A Prompt Templates

In this Section, we provide our prompt templates for our experiments

A.1 Prompt: LLM In-Context-Learning

Below is the prompt template for our few-shot ICL experiments. In this example, we perform 2-shot. Given, the two previous examples are either selected randomly or nearest in the embedding space for the source sentences.

```
### INSTRUCTION:
Translate the input from
English to German.
```

```
###Input: [SRC1]
####Response: [TGT1]
```

```
###Input: [SRC2]
####Response: [TGT2]
```

```
###Input: [SRC3]
####Response:
```

A.2 Prompt: LLM Adapter

We use the following template when adapting the LLM for sentence-level translation. Note that it is different from the ICL in Section A.1. However, we experimented with the below prompt for ICL and found similar results. We do not again perform the experiments for all configurations due to the computational load.

```
[INST] <<SYS>>\nYou are a
professional translator
from English to German.
```

```
The output should only be the
translation in one line.<</SYS>>
```

```
English: [SRC]
[/INST]
German:
```

A.3 Prompt: LLM ZeroShot PE

Since this scenario is zero-shot, we provide more instructions so that is much easier for the model to understand the task. Since we found it was generating explanations and notes even when explicitly

mentioned not to, we ask to always end the answer with "###". Later, we use it as a separator for parsing the output. We provide the prompt template below

```
[INST] <<SYS>>You are a post-editor.
You improve translations from English
to German using the English source and
German translation. Do not provide any
explanation or correction.
The translation should end with
### in new line
<</SYS>>
English: [SRC]
German Translation: [HYP]
[/INST]
Post-Edited Translation:
```

A.4 Prompt: LLM LoRA Sentence APE

In this case, we decrease the prompt size as now we perform fine-tuning and instructions are not necessary. Furthermore, it will lead to less memory consumption as the sequences are much shorter.

```
English: [SRC]
German Translation: [HYP]
Post-Edited Translation: [REF]
```

A.5 Prompt: LLM LoRA Document APE

We format the prompt similarly to the sentence level. The only difference is that now we have sentences separated by "<SS>" token in the document.

```
English: [SRC1] <SS> [SRC2] <SS> [SRC3]
German Translation: [HYP1] <SS> [HYP2]
                  <SS> [HYP3]
Post-Edited Translation: [REF1] <SS>
                        [REF2] <SS> [REF3]
```

B LLM In-Context-Learning: Zero to 5-shot

Please refer to Table 7 for results on ICL with random selection and Table 8. The results reported are on the Must-C v3 test set using LLama2.

C Training Details

C.1 Llama2 Experiments

We use the transformers library (Wolf et al., 2020) for training and inference with Llama2. While training the adapters, we set the hyper-parameters to rank 8, alpha 32, dropout 0.1, and bias as 'LoRA_only'. Following Dettmers et al. (2023) to make the model robust to

Shot Size	BLEU	ChrF	COMET
Sent-level $\triangle$ LM	30.45	57.0	0.8179
0	20.47	48.3	0.7592
1	20.73	48.8	0.7697
2	21.53	50.0	0.7795
3	20.34	50.1	0.7685
4	19.87	50.0	0.7609
5	20.33	50.5	0.7658

Table 7: In-Context-Learning with Llama2 13B using Random Selection Strategy

Shot Size	BLEU	ChrF	COMET
Sent-level $\triangle$ LM	30.45	57.0	0.8179
0	20.47	48.3	0.7592
1	21.16	49.8	0.7755
2	21.13	50.2	0.7724
3	19.61	49.8	0.7593
4	18.82	49.9	0.7531
5	18.51	49.9	0.7402

Table 8: In-Context-Learning with Llama2 13B using FAISS Selection Strategy

LoRA hyper-parameters, we adapt on all layers. The modules we add to the adapter include $q_proj, k_proj, v_proj, gate_proj, up_proj$ and $down_proj$. We set a batch size for each device to 32 initially and enable *auto_find_batch_size* to *True* on 4 NVIDIA RTX A6000 GPU's. To simulate a larger batch size, we set *gradient_accumulation_steps* to 20. We use a *learning_rate* of $2e - 5$. The other parameters are set to default. We train for 3 epochs and select the model with the best validation loss. During inference, we use beam search with a *num_beams* set as 3 as we find it to be reasonable given the computation and performance.

C.2 DeltaLM Experiments

We use the fairseq library (Ott et al., 2019) for our experiments with $\triangle$ LM. During training, we use cross-entropy loss with label smoothing set to 0.1. We set a learning rate of 0.0001 with Adam optimizer, betas (0.9, 0.98) and the initial learning rate to $1e - 7$. We set both dropout and attention dropout to 0.1. We use a batch size of 2000 max tokens and perform gradient accumulation for 3 steps. We train until the validation loss increases after 5 consecutive interval steps that are set to 4500

steps (Roughly 1/3 of epoch). During inference, we do beam size with the number of beams set to 5. The other parameters not mentioned are set to default.

D ContraPro Scores for Sentence and Document APE

	Cxt Size	Contra/Gen (%)
Post and Junczys-Dowmunt (2023)	10	77.9/70.5
Lupo et al. (2023)	4	82.54/_
△ LM + Llama2 13B Sent APE	0	60.0/_
△ LM + Llama2 13B Sent APE	2	85.8/_
△ LM + Llama2 13B Doc APE	0	50.9/_
△ LM + Llama2 13B Doc APE	2	87.7/68.0
△ LM + Llama2 13B Doc APE	4	88.7/69.7

Table 9: Comparing Sentence and Document APE Accuracy on the ContraPro English → German Test Set. For generative results, we only report on sentences from 1 to 10 using the evaluation script from [Post and Junczys-Dowmunt \(2023\)](#).

Source	This is a sentence in Spanish: Las prendas bestsellers se estampan con motivos fLoRAles, animal print o retales tipo patchwork.
Reference	Dies ist ein Satz auf Spanisch: Las prendas bestsellers se estampan con motivos fLoRAles, animal print o retales tipo patchwork.
$\triangle$ LM Hypothesis	Das ist ein Satz auf Spanisch: Die Bestsellers se multidisciplinan conã fLoRAles, Tierdruck oder Reliefs ol Flitterwerk.
Post-Edited with Llama2 13B	Das ist ein Satz auf Spanisch: Las prendas bestsellers se estampan con motivos fLoRAles, animal print o retales tipo patchwork.

Table 10: Example from the ACL dev set taken from Talk id: 268 and Sentence 26. The $\triangle$ LM translates everything into German including the Spanish phrase that needs to be retained in the original language. However, after APE, Llama2 13B does not translate the Spanish Phrase as it was also not translated in the source sentence.

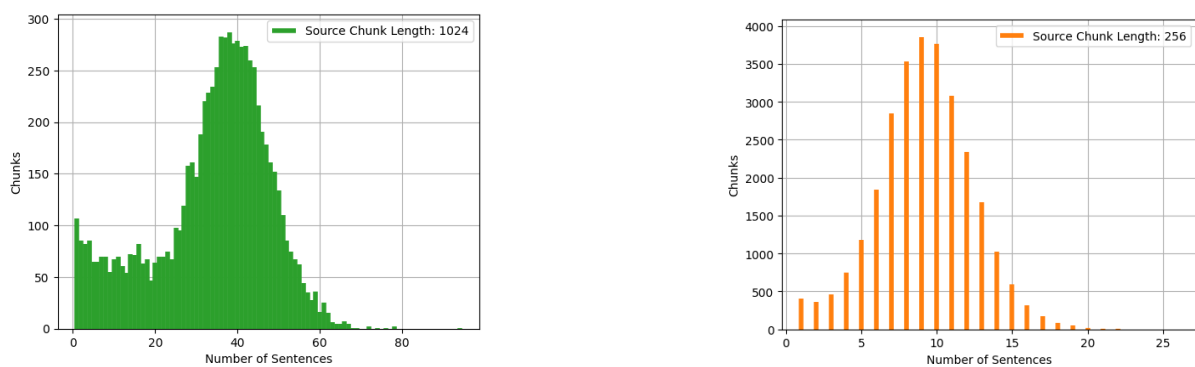


Figure 2: Number of sentences in a document with chunk sizes 1024 and 256.